9. IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER

EX.NO: 9
Date: 8-9-26

Aim:

To design, configure, and implement classless IP subnetting (/27) using cisco packet tracer simulator and verify connectivity b/w different subnets.

Subnetting:

Subnetting is a process of partitioning a single physical network into multiple logical network spaces known as subnets.

Default subnet mask(/24): 255.255.255.0
Borrowed host bits: 3 bits.
New subnet mask(/27): 255.255.255.224
Hosts per subnet $2^5 - 2 = 30$

Procedure:

step 1: Open cisco packet tracer and create.
step 2: Drag 2 routers, switches, PC and connect all connections.

step 3: Divide the network 192.168.1.0/24 into smaller subnets using a /27.

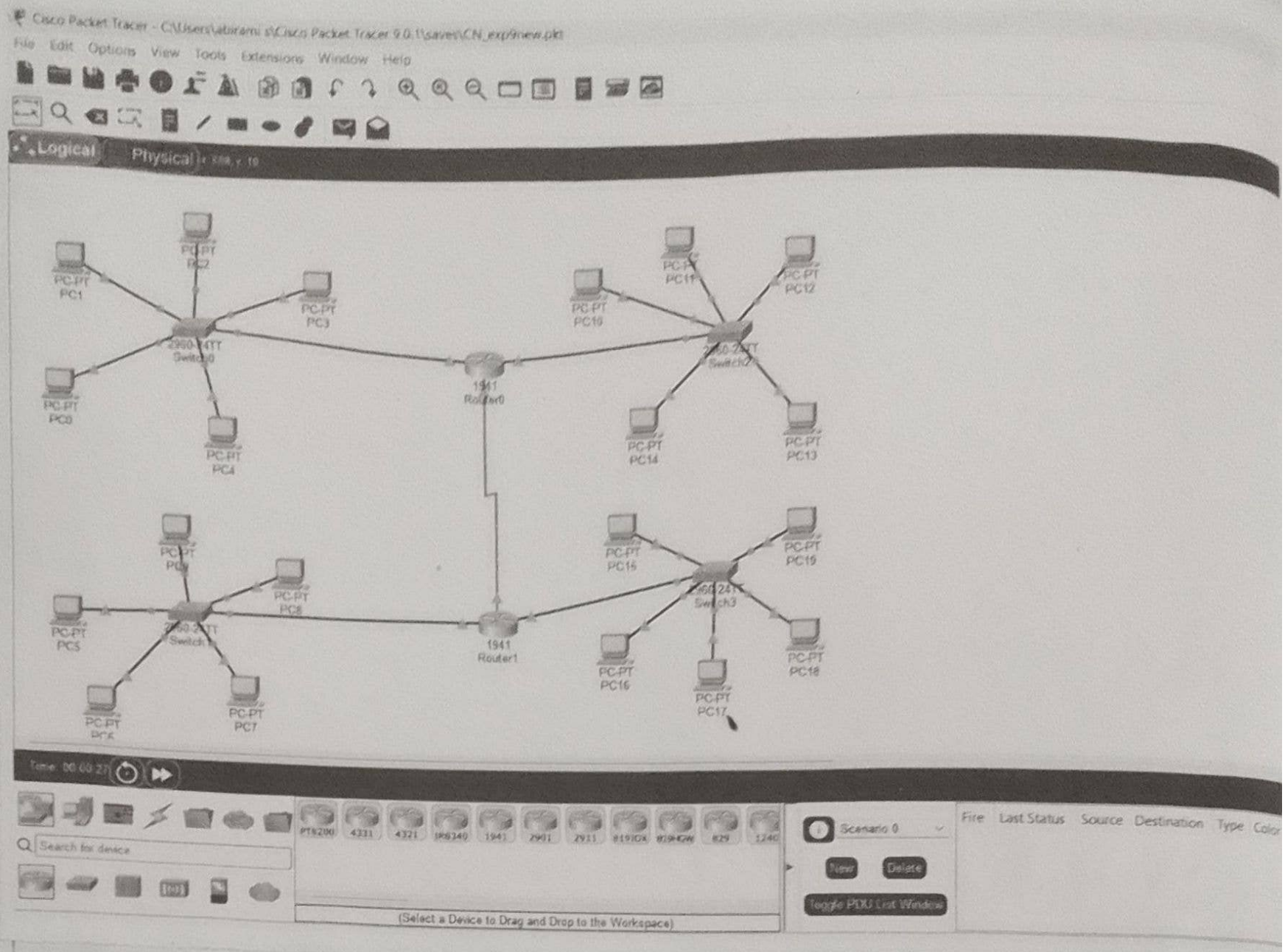
step 4: Configure the router interfaces with the given IP addresses and subnet masks.

step 5: Configure the switch ports according to the given topology.

step 6: Assign IP addresses, subnet masks and default gateways to the PCs.

step 7: Enable the router interfaces using no

step 8: check the network connectivity using the ping.



Result:

Thus, subnetting was implemented configured with appropriate IP addresses and subnet masks. Network connectivity was tested using ping.